

# Spectral Rendering Without a Spectral Renderer: Learned Spectral Codes for RGB Pipelines

JIAQI YU, University of York, United Kingdom

DAR'YA GUARNERA, University of York, United Kingdom

GIUSEPPE CLAUDIO GUARNERA, University of York, United Kingdom and Lumirithmic Ltd, United Kingdom

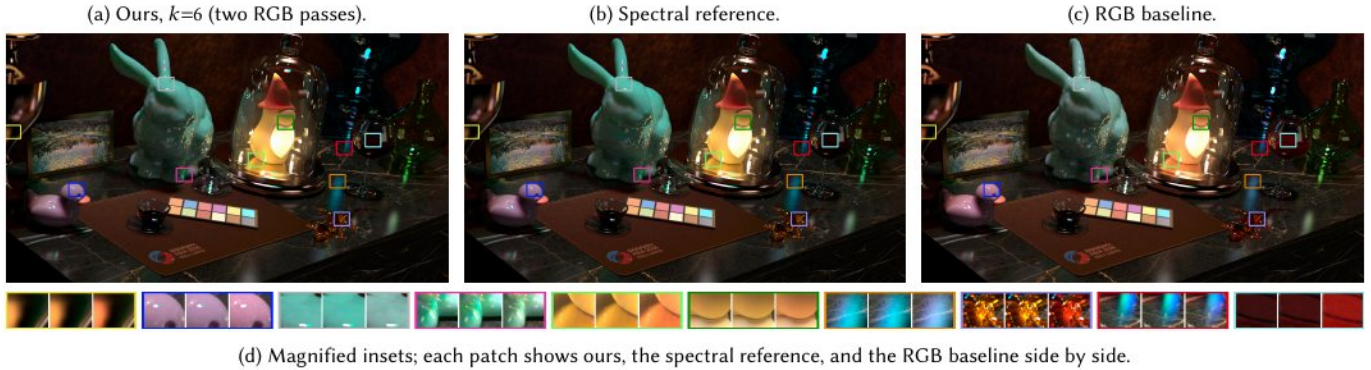


Fig. 1. **A spectral image rendered through an unmodified RGB path tracer.** Our codec (a) approximates the spectral ground truth (b) in two standard RGB passes; the conventional RGB render (c) shows visible chromatic shifts. The insets (d) magnify the regions marked in (a)–(c), rendered at higher sample counts; the three columns of each inset share one exposure, normalised per patch for legibility. The scene exercises every supported material class (spectral textures, measured Schott glass and clear glass, copper and gold conductors, dipole and homogeneous subsurface scattering, and participating-medium fog) under mixed warm/cool broadband illumination with red, green, and blue accents.

Spectral rendering reproduces the wavelength-dependent appearance that RGB rendering cannot: metamerism, colour shifts under spectrally rich illumination, coloured glass, metallic reflection and subsurface scattering. It requires a spectrally aware renderer, but the rendering ecosystem is predominantly RGB: game engines, cloud render farms, GPU rasterisers and most offline renderers consume RGB alone, with no access to the internal wavelength propagation that spectral rendering relies on.

We introduce *learned spectral codes*, a compact linear representation that enables general spectral rendering through *unmodified* RGB pipelines. A trained encoder compresses any visible-range spectral quantity into a small latent vector; the codes are packed as RGB triplets and processed by an off-the-shelf RGB renderer. The codec preserves the arithmetic of light transport, exactly for scaling and addition and approximately for element-wise products, so two standard RGB passes suffice to produce results closely matching a full spectral rendering. The same codec handles reflectance, illumination, coloured glass, conductors, and participating media: one pipeline across material classes, with no per-material modification to the renderer.

Where spectral asset data is unavailable, lightweight neural upsamplers bridge legacy RGB inputs to the latent space. Across diverse materials and illumination, the method yields substantial quality improvements on the

cases where RGB rendering fails most visibly, at rendering cost comparable to efficient spectral-sampling techniques and with no changes to the renderer.

CCS Concepts: • **Computing methodologies** → **Rendering**; *Reflectance modeling*; *Dimensionality reduction and manifold learning*; *Neural networks*.

Additional Key Words and Phrases: spectral rendering, spectral compression, RGB rendering pipelines, multi-pass rendering, spectral upsampling, Hadamard product, metamerism

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## 1 Introduction

Spectral rendering propagates full spectral distributions through the rendering pipeline, reproducing wavelength-dependent phenomena that RGB rendering cannot: metamerism, prismatic dispersion, and thin-film interference [Watanabe et al. 2013]. Modern spectral renderers achieve this efficiently: hero wavelength sampling [Wilkie et al. 2014] and GPU-accelerated spectral path tracers [Murray et al. 2021] bring spectral rendering to near-RGB cost in open-source and custom production pipelines.

Much of the wider rendering ecosystem, however, remains RGB-based. Spectral pipelines such as Manuka [Fascione et al. 2018] exist in production, but commercial game engines, open-source suites such as Blender, GPU rasterisers, cloud render farms, and most proprietary offline renderers expose RGB inputs and RGB frame-buffers only; modifying them to propagate stochastic wavelengths is impractical.

Authors' Contact Information: Jiaqi Yu, [jiaqiyu0307@outlook.com](mailto:jiaqiyu0307@outlook.com), University of York, York, United Kingdom; Dar'ya Guarnera, [darya.guarnera@york.ac.uk](mailto:darya.guarnera@york.ac.uk), University of York, York, United Kingdom; Giuseppe Claudio Guarnera, [claudio.guarnera@york.ac.uk](mailto:claudio.guarnera@york.ac.uk), University of York, York, United Kingdom and Lumirithmic Ltd, London, United Kingdom.



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Recent work delivers individual spectral effects without touching the renderer core: Tominaga et al. [2026] render fluorescence through Mitsuba 0.6's spectral build by encoding the re-radiation into per-channel inputs, and Fichet et al. [2024] obtain a similar effect with a reduced re-radiation matrix implemented at the shader level. Fluorescence itself lies outside our supported regime, so these methods are complementary to ours.

Rendering general spectral content through multiple passes of an RGB pipeline originates with Darling et al. [2011], who relight scenes with a fixed six-channel parametric basis. We generalise this assets-only route, which requires no custom code anywhere: a single representation for per-channel-preserving spectral transport across reflectance, illumination, glass, conductors, and participating media, rendered through entirely unmodified RGB renderers. This representation, a *learned spectral code*, is a compact latent vector  $z \in \mathbb{R}^k$  that supports the per-channel arithmetic an RGB renderer already performs: scaling, addition, and element-wise multiplication. Spectral quantities encode into  $z$ ; the codes pack into  $k/3$  RGB triplets; the renderer processes them in  $k/3$  unmodified passes; a final decode recovers the spectral radiance (Fig. 2 illustrates the full pipeline). Whether the renderer is Mitsuba, Blender Cycles, Maya Arnold, KeyShot, or a real-time rasteriser (Unreal Engine 5) is immaterial: each processes RGB triplets per channel, which is what the codec requires.

Exact preservation of all three operations is impossible whenever the code dimension  $k$  is smaller than the number  $n$  of spectral bins (Sec. 3.2), so the encoder is approximate by necessity. Our linear, non-negative architecture preserves scaling and addition *exactly* and is trained to approximate element-wise multiplication on the manifold of natural visible spectra. Since each RGB pass carries three channels, practical operating points are multiples of three; Sec. 6.1 establishes  $k=6$  as the smallest configuration that is meaningfully more accurate than a standard RGB pipeline, with  $k=3$  blocked by a Luther–Hadamard trade-off (alignment with the colour-matching subspace [Ives 1915; Luther 1927] against Hadamard preservation) and  $k=9$  unwarranted in quality terms.

The codec is content-agnostic: the same encoder–decoder embeds reflectance, illumination, transmittance, conductor IOR, and absorption and scattering coefficients into the same latent space (Sec. 4); two per-pixel upsamplers project legacy RGB assets onto the codec's manifold (Sec. 5). Our scope covers wavelength-preserving interactions; wavelength-shifting effects (fluorescence, dispersion) lie outside the supported regime (Sec. 7).

#### Contributions.

- (1) A **learned non-negative spectral basis and its dual**, jointly optimised so that iterated per-channel products of the resulting  $k$ -dimensional codes remain accurate. This codec compresses any visible-range spectral quantity and enables general spectral rendering through unmodified RGB renderers in  $k/3$  standard passes (Sec. 4).
- (2) **Latent upsamplers** for legacy RGB assets that target the codec's  $k$ -dimensional manifold rather than full  $n$ -channel spectra (Sec. 5).
- (3) A **capacity analysis** establishing  $k=6$  as the practical operating point:  $k=3$  blocked by a Luther–Hadamard trade-off,  $k=9$

unwarranted in quality terms, and the learned basis irreplaceable at  $k=6$  (Sec. 6.1).

We evaluate the codec on broadband, narrowband, and multi-peaked illumination across surface reflectance, coloured-glass transmission, conductor reflection, and homogeneous and heterogeneous subsurface scattering; the teaser (Fig. 1) exercises all of these in one scene. We benchmark timing and reconstruction quality against Mitsuba 3's hero wavelength sampling at matched resolution and spp, and we deploy the codec across four production renderers (Sec. 6).

## 2 Related Work

### 2.1 Compact spectral representations

Most compact spectral representations, ours included, are linear decompositions. Peercy [1993] introduced low-dimensional point-sampled representations for spectral rendering; PCA has been applied to reflectance and illumination datasets [Otsu et al. 2018] and to measured material capture [Guarnera et al. 2017; Matusik 2003]. PCA admits negative entries; NMF [Berry et al. 2007] is non-negative but needs more dimensions. Both serve as baselines in Sec. 6.1.

Parametric representations replace the data-fitted basis with a constrained family. Peters et al. [2019] encode bounded spectra as Fourier moments with a maximum-entropy reconstruction, achieving 3–8 coefficients with smooth, energy-conserving output. Karimipour et al. [2023] represent reflectance as  $3 \times 3$  matrices fitted to cone responses, reaching  $\Delta E_{94} < 1$  under broadband illumination but degrading to  $\Delta E_{94}$  up to 16 under narrowband sources. Both compress *individual spectra* accurately but provide no algebra on the compressed codes: the product of two compressed spectra cannot be evaluated without decompressing first.

### 2.2 Efficient transport in spectral renderers

Hero wavelength sampling [Wilkie et al. 2014] carries one stochastically chosen wavelength per ray with several stratified companions, reducing colour noise at near-RGB cost. West et al. [2020] generalise wavelength importance sampling to a continuous multiple importance sampling framework, and van de Ruit and Eisemann [2021] build coarse screen-space estimates over multiple passes to guide subsequent sampling. These methods improve efficiency *within* a spectral renderer; we target renderers with no spectral path at all.

### 2.3 RGB-to-spectrum upsampling

Converting RGB textures to plausible spectra is a long-standing problem. Smits [1999] introduced a scaled basis-spectrum conversion with a long production history: the Manuka renderer relied on a Smits-style optimisation for all its uplifting [Fascione et al. 2018], and recent versions replace it with the moment-based method of Peters et al. [2019] for reflectances [Weidlich et al. 2021]. Meng et al. [2015] introduced a physically meaningful uplift over the XYZ space. Jakob and Hanika [2019] parameterise the sRGB gamut with three coefficients, smooth and energy-conserving; the method is now standard in PBRT-v4 [Pharr et al. 2023] and Mitsuba 3 [Jakob et al. 2022]. Jung et al. [2019] uplift gamuts wider than sRGB by adding fluorescent components. Belcour et al. [2023] give metameric one-to-many uplift for textured assets. Tódová et al. [2021; 2022] constrain the uplift so that selected spectra are reproduced exactly;

Tódová and Wilkie [2025] extend uplifting to HDR environment maps, and van de Ruit and Eisemann [2024] constrain uplifted spectra by their rendered RGB appearance, including indirect illumination. Guarnera et al. [2022] target illumination spectra specifically, uplifting a light’s tristimulus value by optimising blending weights over a compact spectral basis. They propose two variants: a measured basis of six LED primaries, which produces SPDs with the spectral structure of real multi-primary sources, and a smooth parametric Gaussian basis. Both reproduce the target chromaticity by construction; as with reflectance uplifting, the reconstructed SPD is one metamer of many, and the choice of basis decides which one. Xia et al. [2025] sample the null space of the RGB constraint via differentiable rendering. We benchmark our reflectance upsampler against Jakob and Hanika [2019] and our illumination upsampler against Guarnera et al. [2022] in Sec. 6.3. While these methods uplift RGB to  $n$ -channel spectra rendered with a spectral renderer, ours map RGB to  $k$ -dimensional latent codes rendered with an RGB renderer.

## 2.4 Spectral effects in RGB-only renderers

A small but growing line of work delivers spectral phenomena without a spectral renderer. Ward and Eydelberg-Vileshin [2002] prefilter materials and lights before rendering, achieving near-exact direct illumination at the cost of approximate interreflection. Sun [2006] decomposes thin-film iridescence into a small number of RGB passes through an unmodified renderer. Fichet et al. [2024] handle fluorescence by replacing the albedo–light product with a reduced re-radiation matrix, requiring shader-level changes; their reduced-matrix formalism itself is general, and we adopt it in Sec. 3.4 to analyse our codec’s constraints. Each of these methods is constructed for a specific phenomenon. We take a different route: a single learned codec supports general *wavelength-preserving* spectral transport (reflectance, illumination, coloured glass, conductors, and homogeneous and heterogeneous participating media) through any RGB renderer with no engine modification. Wavelength-shifting effects (fluorescence, phosphorescence, dispersion) couple distinct wavelengths during transport and remain outside the per-channel regime our codec preserves.

## 3 Learned Spectral Codes

Fig. 2 summarises the system: a jointly trained encoder–decoder pair around otherwise unmodified RGB render passes. This section develops the constraints that shape it.

### 3.1 Algebraic constraints from light transport

Path tracing repeatedly applies three operations to spectral quantities: scaling (intensity, BSDF evaluation), addition (multiple light contributions, path accumulation), and *element-wise* (Hadamard multiplication: reflectance with incident radiance, transmittance with through-flux, per-channel attenuation through participating media, and conductor Fresnel evaluated at the complex index  $\eta(\lambda) - i\kappa(\lambda)$ ).<sup>1</sup> Discretising the visible spectrum on  $n$  wavelength bins yields the algebra  $\mathcal{A} = (\mathbb{R}^n, +, \odot)$  on which transport acts.

<sup>1</sup>We use  $\kappa$  for the imaginary part of the conductor index to avoid collision with the latent dimension  $k$ .

A renderer-friendly code must support the same three operations *directly on the compact codes*, so that the renderer can treat codes as ordinary  $k$ -channel inputs. Formally, this requires the encoder  $E$  to act as a ring homomorphism  $\mathcal{A} \rightarrow (\mathbb{R}^k, +, \odot)$ .

### 3.2 Impossibility of exact algebra-preserving compression

The compression we seek does not exist exactly:

**THEOREM 3.1.** *When  $k < n$  there is no injective ring homomorphism  $\phi : (\mathbb{R}^n, +, \odot) \rightarrow (\mathbb{R}^k, +, \odot)$ .*

The argument is that the standard basis  $\{e_1, \dots, e_n\}$  forms a system of  $n$  pairwise orthogonal nonzero idempotents under  $\odot$ , while  $(\mathbb{R}^k, +, \odot)$  admits at most  $k$  such idempotents [Atiyah and Macdonald 1969; Lang 2002], so for  $k < n$  at least one  $e_i$  lies in  $\ker \phi$ . The full proof and algebraic development are in Supplemental Sec. 1.

The natural relaxation is *distributional* preservation: scaling and addition are preserved exactly (a property our linear architecture below guarantees by construction), while Hadamard multiplication is preserved approximately on the low-dimensional manifold of natural visible spectra learned from data. The asymmetry is deliberate. Scaling and addition enter every radiance estimate through path accumulation and Monte-Carlo averaging, where a systematic error would bias the estimator itself; the Hadamard product enters once per material interaction, where a small approximation error is perceptually tolerable, and the multi-bounce measurements in Sec. 6.2 show that it does not accumulate.

### 3.3 Linear non-negative codec

We instantiate the encoder  $E$  and decoder  $D$  as linear maps

$$z = E(s) = W_{\text{enc}} s, \quad \hat{s} = D(z) = W_{\text{dec}} z, \quad (1)$$

with  $W_{\text{enc}} \in \mathbb{R}^{k \times n}$  and  $W_{\text{dec}} \in \mathbb{R}^{n \times k}$ . Linearity gives  $E(\alpha s) = \alpha E(s)$  and  $E(s_1 + s_2) = E(s_1) + E(s_2)$  exactly; only the Hadamard property is learned.

Both weight matrices are constrained non-negative through a softplus reparameterisation of unconstrained learnable matrices.<sup>2</sup> Non-negativity is essential, not cosmetic: codes of non-negative spectra remain non-negative, so they pack into RGB framebuffers and flow through an unmodified forward-light-only renderer. The columns of  $W_{\text{dec}}$  admit a direct interpretation as a basis of spectra, with the latent code being the non-negative mixture-weight vector for that basis.

### 3.4 The codec as a diagonal reduced matrix

Fichet et al. [2024] describe spectral reduction by a pair of maps: an analysis map (our  $W_{\text{enc}}$ ) and a reconstruction basis (our  $W_{\text{dec}}$ ). In their framework, a material whose spectral action is the diagonal operator  $\text{diag}(m)$ , with  $m \in \mathbb{R}_{\geq 0}^n$  the material spectrum, acts on the reduced coefficients through the  $k \times k$  *reduced matrix*

$$R_m = W_{\text{enc}} \text{diag}(m) W_{\text{dec}},$$

$$(R_m)_{ij} = \sum_{\lambda} W_{\text{enc}}[i, \lambda] m(\lambda) W_{\text{dec}}[\lambda, j]. \quad (2)$$

<sup>2</sup>Parameterisation details, signed-decoder ablations, basis visualisations, and fidelity diagnostics are in Supplemental Sec. 2.

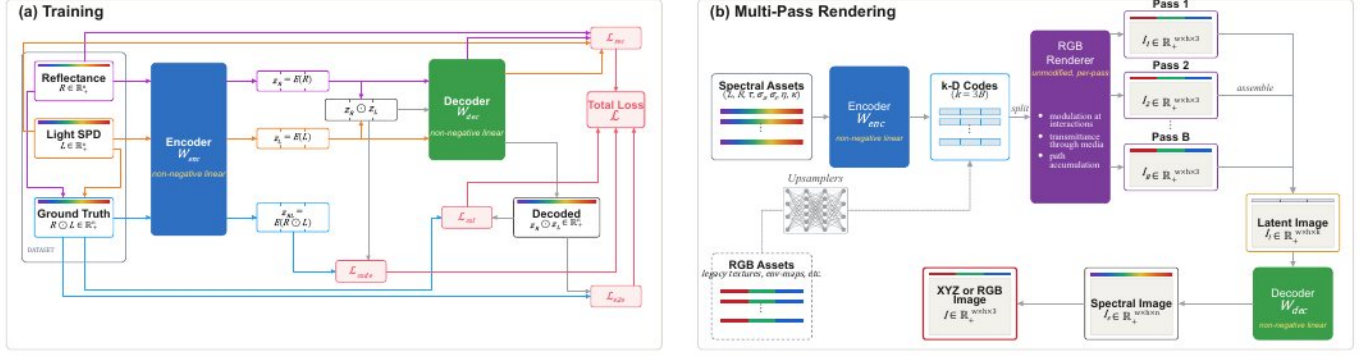


Fig. 2. **Learned spectral codec and multi-pass rendering pipeline.** (a) **Training:** a non-negative encoder  $E (W_{enc} \in \mathbb{R}_+^{k \times n})$  and decoder  $D (W_{dec} \in \mathbb{R}_+^{n \times k})$  are fit jointly on reflectance and illumination spectra; the same matrices embed transmittance  $\tau$ , absorption  $\sigma_a$ , scattering  $\sigma_s$ , and conductor IOR  $\eta, \kappa$  at inference. Four loss terms ( $\mathcal{L}_{2e}$ ,  $\mathcal{L}_{rec}$ ,  $\mathcal{L}_{code}$ ,  $\mathcal{L}_{col}$ ) are combined (Sec. 3.6). (b) **Rendering:** spectral assets encode once into  $k$ -dimensional codes ( $k=3B$ ); legacy RGB assets bypass the encoder via per-pixel upsamplers  $f^{(R)}, f^{(L)}$  (dashed). The code is split into  $B$  RGB-format triplets, processed by  $B$  unmodified RGB passes, then reassembled and decoded;  $B=2$  at  $k=6$ .

Applying general reduced matrices requires shader-level matrix-vector products, which is exactly the modification Fichet et al. make. An unmodified RGB renderer can only multiply per channel, so it applies  $\text{diag}(z_m)$ : our setting is the restriction of the reduced-matrix framework to *diagonal*  $R_m$ .

Requiring  $(R_m)_{ij} = 0$  for all  $i \neq j$  and for every non-negative material spectrum  $m$  forces each summand in Eq. (2) to vanish:  $W_{enc}[i, \lambda] W_{dec}[\lambda, j] = 0$  at every  $\lambda$ , i.e. encoder row  $i$  and decoder column  $j$  must have disjoint supports for  $i \neq j$ . Non-negativity is what makes this rigid: with signed matrices the off-diagonal integral could cancel across wavelengths without the supports being disjoint, whereas non-negative summands admit no cancellation. At the same time, a faithful round-trip requires each encoder row to overlap its *own* decoder column. Exact diagonality for all  $m$  therefore forces a partition of the spectrum: each basis function owns a wavelength range, which is precisely binning.

Binning, however, caps accuracy: the comparison in Sec. 6.1 places every pure-binning variant at roughly  $3 \times$  our error. The codec relaxes exact diagonality to diagonality *in distribution*: the training objective penalises off-diagonal leakage only on products of natural spectra, not for every conceivable  $m$ . This relaxation admits learned solutions that binning excludes, and the learning is where the accuracy is earned: in the same comparison, replacing the bin-average encoder with a learned one nearly halves the error of fixed boxes, and learning the decoder jointly nearly halves it again. The learned matrices keep nearly disjoint supports with small, deliberate overlaps (Supplemental Sec. 2). The codec is best understood as the distributionally-diagonal member of the reduced-matrix family that minimises rendered colour error.

The codec is *content-agnostic*: the same  $E, D$  encode reflectance  $R$ , illumination  $L$ , transmittance  $\tau$ , absorption  $\sigma_a$  and scattering  $\sigma_s$  coefficients, and the spectral parts of the conductor IOR  $\eta, \kappa$  into a common latent space. Fig. 3 shows the learned matrices that realise these constraints: smooth, band-localised decoder columns with the near-disjoint supports derived above, and sparser, multi-peaked encoder rows acting as band-pass analysers.

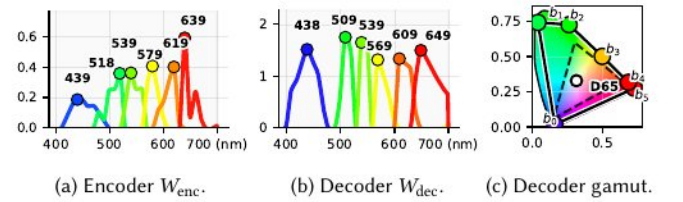


Fig. 3. **Learned codec basis.** Decoder columns (b) are smooth, band-localised, and span a chromaticity gamut (c) larger than sRGB on the green/blue/red edges. Encoder rows (a) are sparser and multi-peaked, acting as band-pass analysers rather than the transpose of the decoder. Channels are ordered by decoder peak wavelength; the same index links the curves of (a) and (b) and the vertices  $b_i$  of (c).

### 3.5 Pass packing

We choose  $k=3B$  so the latent code splits into  $B$  blocks of three channels. Each block is a triplet; the renderer treats it as an ordinary RGB colour, applies its standard per-channel operations, and writes the output to the corresponding block of an HDR framebuffer. The full  $k$ -dimensional code is processed in  $B$  unmodified RGB passes. For two spectra  $s_R, s_L$  encoded as  $z_R, z_L \in \mathbb{R}^k$  the renderer computes

$$z_R \odot z_L = \bigoplus_{b=1}^B (z_R^{(b)} \odot z_L^{(b)}), \quad (3)$$

with  $z^{(b)} \in \mathbb{R}^3$  the  $b$ -th triplet and  $\bigoplus$  concatenation. Eq. (3) introduces no new operation: it is the same  $\odot$  on a  $k$ -vector, written to expose the renderer's per-pass three-channel structure. The training objective, defined next, drives the codec toward

$$E(s_R \odot s_L) \approx E(s_R) \odot E(s_L), \quad (4)$$

on the data manifold, so the per-pass product the renderer computes matches the encoded product of the spectral interaction.

### 3.6 Training objectives

The codec is trained on a corpus of measured and synthetic reflectance spectra (Munsell extended to sRGB-edge coverage) and an augmented illumination corpus (LSPDD-measured lamps [LSPDD 2026], CIE daylight, and synthetic narrowband variants, prefiltered by bin integration so that mercury lines and other narrow peaks survive downsampling). Full corpus construction, split methodology, and representative test spectra are in Supplemental Sec. 3.3.

The codec minimises

$$\mathcal{L} = \lambda_{e2e} \mathcal{L}_{e2e} + \lambda_{rec} \mathcal{L}_{rec} + \lambda_{code} \mathcal{L}_{code} + \lambda_{col} \mathcal{L}_{col}, \quad (5)$$

each term targeting one component of the spectral pipeline:

$$\begin{aligned} \mathcal{L}_{e2e} &= \Phi(D(E(R) \odot E(L)), R \odot L) \\ \mathcal{L}_{rec} &= \text{MSE}(\text{DER}, R) + \text{MSE}(\text{DEL}, L) \\ \mathcal{L}_{code} &= \text{MSE}(E(R) \odot E(L), E(R \odot L)) \\ \mathcal{L}_{col} &= \Delta E_{76}(D(E(R) \odot E(L)), R \odot L). \end{aligned}$$

$\mathcal{L}_{e2e}$  is the end-to-end fit between the codec-rendered product and the spectral ground truth, with  $\Phi(x, y) = (2 - \cos(x, y)) \text{MSE}(x, y)$  combining magnitude and shape.  $\mathcal{L}_{rec}$  enforces round-trip accuracy on individual spectra.  $\mathcal{L}_{code}$  is the Hadamard-multiplicativity term that Theorem 3.1 forbids from reaching zero in general but that we minimise in distribution.  $\mathcal{L}_{col}$  evaluates the same product-ground-truth pair as  $\mathcal{L}_{e2e}$ , but perceptually: as  $\Delta E_{76}$  in CIELAB under the per-sample illuminant white point.  $\mathcal{L}_{code}$  alone admits the trivial collapse  $E \equiv 0$ ;  $\mathcal{L}_{e2e}$  and  $\mathcal{L}_{rec}$  make this infeasible by requiring the codec to reproduce both individual spectra and their products. The same argument pins the overall scale: rescaling  $E \rightarrow sE$  and  $D \rightarrow D/s$  leaves  $\mathcal{L}_{rec}$  unchanged, but  $\mathcal{L}_{e2e}$  and  $\mathcal{L}_{col}$  apply the encoder twice and the decoder once, so their error grows as  $s$  shrinks and the degeneracy is excluded. Non-negativity is enforced architecturally by the softplus parameterisation and needs no penalty.

Eq. (5) trains on single reflectance-illuminant products; it is  $\mathcal{L}_{code}$  that trains the latent product structure which iterates over multiple bounces, and multi-bounce accuracy is measured empirically (Sec. 6.2). The corpus contains reflectance and illumination spectra only; transmittance, complex IOR, and scattering coefficients are encoded with the same matrices, without retraining (Sec. 4).

The four weights  $\lambda_\bullet$  were selected by a two-phase random search over 130 codec variants, optimising for stability under chained products (three or more spectra, i.e. multiple rendering bounces); the search ranked candidates by  $\Delta E_{94}$  for speed, while all reported results use  $\Delta E_{2000}$ . The selected values, the search protocol, and the multi-bounce sensitivity sweep are in Supplemental Sec. 3.2.

### 4 Material-Class Encoding

The codec compresses any visible-range spectral quantity the same way (Sec. 3.1): whichever spectral input a BSDF or volume model expects (reflectance, transmittance, complex index of refraction, absorption or scattering coefficients) is encoded once into a  $k$ -dimensional latent code and processed by the unmodified renderer in  $B=k/3$  standard RGB passes. The renderer’s per-channel arithmetic (scaling, addition, Hadamard products at material interactions, and per-channel attenuation through participating media) operates on the  $k$  latent channels exactly as it would on three RGB channels.

*Coloured glass.* Coloured glass is characterised by a wavelength-dependent transmittance  $\tau(\lambda) \in [0, 1]$  that attenuates light passing through it. We handle two physically distinct regimes. For thin filters and surface coatings, the transmittance acts as a per-crossing multiplier on the radiance: at each interface  $L_{out}(\lambda) = \tau(\lambda) L_{in}(\lambda)$ . Encoding  $z_\tau = E(\tau)$  once and applying  $z_\tau^{(b)}$  per channel across  $N$  surface crossings corresponds to integer Hadamard powers  $z_\tau^N$ , the iterated-product structure the codec is trained to approximate. For bulk-absorbing glass, spectral attenuation follows the Beer–Lambert law along the optical path:  $T(d, \lambda) = \exp(-\sigma_a(\lambda) d)$ . We obtain a latent absorption coefficient by inverting the encoded transmittance at a reference depth,  $\sigma_a^{(b)} = -\ln(z_\tau^{(b)})/d_{ref}$ ; the renderer’s per-channel volume integrator then applies  $\exp(-\sigma_a^{(b)} d)$  along each ray segment of length  $d$ , a fractional Hadamard power the codec approximates.

*Conductors.* A conductor BSDF is parameterised by a wavelength-dependent complex index of refraction  $\eta(\lambda) - i\kappa(\lambda)$ . We encode  $\eta$  and  $\kappa$  as *ordinary spectra* through the same codec, recovering two  $k$ -dimensional codes  $z_\eta, z_\kappa$ , and feed them per-pass to the unmodified conductor BSDF. The Fresnel equations are non-linear in  $\eta, \kappa$ , but the non-linearity acts *per channel*: the renderer evaluates Fresnel on the  $k$  latent channels just as it would on three RGB channels.<sup>3</sup> Replacing one accurate spectral Fresnel evaluation with  $k$  approximate per-channel ones is far less lossy than the standard RGB collapse because the  $k$  near-disjoint latent channels sample  $\eta$  and  $\kappa$  much more finely than three broad RGB primaries. The renderer remains an opaque RGB BSDF evaluator; the spectral information lives in the latent channels alone.

*Homogeneous participating media.* Absorption  $\sigma_a(\lambda)$  and scattering  $\sigma_s(\lambda)$  coefficients are ordinary non-negative visible-range spectra, which the codec embeds directly. We encode each separately, and the renderer’s standard per-channel volume integrator (the same code path it uses for an RGB homogeneous medium) accumulates per-channel transmittance  $\exp(-(\sigma_a^{(b)} + \sigma_s^{(b)}) d)$  and applies the single-scatter albedo  $\sigma_s^{(b)} / (\sigma_a^{(b)} + \sigma_s^{(b)})$  as a per-channel multiplier at scattering events, the transport that produces subsurface scattering (SSS) in translucent materials. Media therefore exercise two per-channel regimes: extinction follows the Beer–Lambert analysis above, and the scattering albedo is a per-channel ratio of two encoded spectra. The same codec, trained on reflectance and illumination only, encodes extinction coefficients without retraining. For media with strongly saturated absorption bands, whose albedo the ratio regime cannot represent accurately, we factor the transmittance into a per-crossing chromatic term, encoded as a surface transmittance, and a smooth residual absorption coefficient carried by the medium.

*Heterogeneous participating media.* Per-voxel spectral data is encoded into the same  $k$ -dimensional latent space and stored as  $B$  RGB-format grid volumes; each pass reads its triplet through the renderer’s heterogeneous-medium machinery without modification. Linearity of the encoder makes this both practical and exact. Voxel spectra are typically blends of a few feature spectra, and encoding

<sup>3</sup>Per-channel Fresnel evaluation on RGB primaries is known to distort metal colours [Hoffman 2019]; the encoding applies equally when the renderer uses the Schlick approximation.

commutes with blending: the feature spectra are encoded once and their latent codes blended per voxel, so no per-voxel encoding is needed, and the renderer's volume filtering commutes with the encoding for the same reason (exactly so for trilinear and any non-negative linear filter). Along a ray, the accumulated optical depth is a per-channel sum of voxel contributions, and sums are preserved exactly, as is rescaling by a scalar density field, which multiplies all channels by the same factor. Heterogeneity therefore introduces no approximation beyond the homogeneous regimes above: the deviation remains that of the per-channel exponential and the albedo ratio.

Per-class encoding parameters and renderer bindings are documented in Supplemental Sec. 6.

## 5 Latent Upsampling for Legacy RGB Assets

While our codec compresses spectral inputs end-to-end, production pipelines hold large libraries of RGB textures, materials, and environment maps for which spectral data is unavailable. The obvious route for such assets is to uplift them with an existing RGB-to-spectrum method, for example Jakob and Hanika [2019], and encode the result. This composes two independent approximations: the uplift selects one metamer with no regard for the codec, and the encoder then compresses a spectrum that need not lie near its training distribution. We instead train lightweight networks that map the RGB input directly to a latent code. The prediction target is the code the codec itself would produce for the underlying spectrum, so the networks learn to land on codes that render accurately: on held-out data the composed route yields about  $1.6\times$  the colour error of direct prediction (the through-codec baseline rows of Tab. 5), while direct prediction stays close to the codec's own accuracy on ground-truth spectra. The upsamplers are a convenience bridge for RGB-only assets, not a spectral reconstruction method.

Both networks map a colour input  $\mathbf{c} \in \mathbb{R}^3$  to a  $k$ -dimensional latent code:

$$\mathbf{z}_{\text{pred}} = \begin{cases} f_{\theta}^{(R)}(\mathbf{c}) & \text{(reflectance assets)} \\ f_{\phi}^{(L)}(\mathbf{c}) & \text{(illumination assets).} \end{cases} \quad (6)$$

The networks operate in linear XYZ: at training time the input is the CMF projection of the training spectrum, and at render time RGB assets are first linearised and converted with the standard sRGB-to-XYZ matrix. We use two networks rather than one because reflectance distributions are bounded ( $s \in [0, 1]^n$ ) and broadly Munsell-like, whereas illumination is unbounded and includes narrowband emission peaks; the two need different inductive biases. Both networks are per-pixel multilayer perceptrons with scale equivariance (the input magnitude is factored out, the network maps the unit direction to a unit-scale latent, and the magnitude is reapplied at the output) so the codec's exact scale equivariance carries through to RGB inputs. The reflectance network additionally has an *auxiliary spectrum head*, a small MLP that decodes the predicted latent back to a reconstructed spectrum, supervised by a smoothness penalty during training. This auxiliary path acts as a regulariser tied to the smooth Munsell-like prior of reflectance and improves reflectance accuracy by  $2.5\times$  (Supplemental Sec. 7); we find no equivalent benefit for illumination, whose narrowband peaks penalise smoothness,

so  $f^{(L)}$  uses a single latent head. Codec weights are frozen during upsampler training.

*Training objective.* For each spectrum  $s$  in the codec's training set the input is the CMF projection  $\mathbf{c} = T_{\text{cmf}} s$  and the target latent is  $\mathbf{z}_{\text{gt}} = E(s)$ . Both networks share a common latent-supervision loss

$$\begin{aligned} \mathcal{L}_{\text{lat}} = & \text{MSE}(\mathbf{z}_{\text{pred}}, \mathbf{z}_{\text{gt}}) \\ & + \lambda_{\text{maxabs}} \mathbb{E}[\|\mathbf{z}_{\text{pred}} - \mathbf{z}_{\text{gt}}\|_{\infty}] \\ & + \lambda_{\text{col}} \Delta E_{76}(\mathbf{D}(\mathbf{z}_{\text{pred}}), \mathbf{D}(\mathbf{z}_{\text{gt}})), \end{aligned} \quad (7)$$

where the MSE enforces overall latent accuracy, the  $\ell_{\infty}$  term controls outlier dimensions, and the perceptual  $\Delta E_{76}$  term, computed on the decoded spectra in CIELAB, ensures colorimetric stability after decode. The reflectance network additionally supervises its spectrum-head output  $\hat{s}_{\text{pred}}$  against the ground-truth spectrum:  $\mathcal{L}_{\text{spec}}^{(R)} = \text{MSE}(\hat{s}_{\text{pred}}, s) + \lambda_{\text{smooth}} \sum_i (\hat{s}_{i+1} - \hat{s}_i)^2$ , yielding the total reflectance loss  $\mathcal{L}^{(R)} = \mathcal{L}_{\text{lat}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}}^{(R)}$ . Loss weights and training schedule are in Supplemental Sec. 7.

*Scope.* Three boundaries delimit what these networks do. First, RGB-to-spectrum upsampling is one-to-many: a colour corresponds to many metamers, and a deterministic network selects a single one. Like all one-to-one upsamplers, ours does not address metamerism; when an asset's true spectrum differs from the selected metamer, relighting can diverge, and smooth selections are the safer choice. We therefore claim no recovery of true spectra: the target is a latent code that reproduces the asset's colour and renders robustly across illuminants. Second, unlike methods that enforce the input colour as a hard constraint, our round-trip is soft: Eq. (7) penalises but does not eliminate colour drift. The cost is small on reflectances (Supplemental Sec. 7) and is most visible under broadband illumination upsampling, where a hard-constrained uplift is close to exact by construction (Tab. 5); in exchange, the soft formulation is free to trade exact colour reproduction at one condition for rendering accuracy across conditions. Third, scale equivariance is the correct prior for illumination, whose overall intensity is arbitrary, but for reflectance it forgoes the brightness information that recent uplift methods exploit to constrain the feasible metamer set; we accept this trade for a single shared mechanism across both asset types.

*Application: spectral assets from RGB inputs.* At render time, RGB textures and environment maps are upsampled per pixel into a  $k$ -dimensional latent map, packed as  $k/3$  RGB-format EXR textures, and rendered through the unmodified renderer (Sec. 6.3).

## 6 Experiments

We evaluate the codec against spectral rendering as ground truth (Mitsuba 0.6 spectral build) and against standard RGB rendering as the practical baseline (the same Mitsuba 0.6 in RGB mode; the spectral build is never touched by our pipeline). We use  $k=6$  (two RGB passes) throughout. All qualitative figures decode latents to spectra and CMF-integrate to sRGB. Sample budgets vary per scene class to balance Monte-Carlo noise against render time and are stated per figure; further tables and ablations are in the supplemental material.

Table 1. **Cornell-box mean  $\Delta E_{2000}$  per illuminant for  $k=6$  representations.** Pure binning uses a bin-average encoder and box decoder with no training; the adaptive variant optimises the bin edges on the training illuminants; the “bins at decoder transitions” variant places bin edges at our learned decoder’s own transition wavelengths. All rows share the pipeline and ground truth of Fig. 4, at 1024 spp.

| $k=6$ representation            | broad <sub>12</sub> | broad <sub>63</sub> | narrow <sub>525</sub> | narrow <sub>351</sub> | FL11        | avg         |
|---------------------------------|---------------------|---------------------|-----------------------|-----------------------|-------------|-------------|
| equal-width bins                | 2.02                | 2.18                | 1.73                  | 10.47                 | 5.36        | 4.35        |
| bins at decoder transitions     | 1.86                | 1.98                | 2.92                  | 9.23                  | 5.00        | 4.20        |
| adaptive-width bins             | 1.98                | 2.07                | 1.33                  | 6.69                  | 5.57        | 3.53        |
| RGB baseline ( $k=3$ )          | 3.40                | 2.42                | 2.89                  | 3.82                  | 4.01        | 3.31        |
| equal boxes, learned encoder    | 1.25                | 2.03                | 2.94                  | 1.42                  | 4.14        | 2.36        |
| Gaussian basis, learned encoder | 0.92                | 1.81                | 1.50                  | 1.10                  | 7.52        | 2.57        |
| ours, jointly learned           | <b>0.90</b>         | <b>1.52</b>         | <b>1.11</b>           | <b>0.86</b>           | <b>1.83</b> | <b>1.24</b> |

### 6.1 Codec reconstruction quality

*Cornell-box evaluation under five illuminants.* We first evaluate the codec at  $k=6$  on a Cornell box containing diverse BRDFs and a ColorChecker chart, rendered under broadband, narrowband, structured, and multi-peaked illumination (Fig. 4; per-condition values in the figure). Materials and illuminants here are spectral ground truth encoded directly through the codec, with no upsampling involved. Across all five illuminants the latent pipeline reduces  $\overline{\Delta E}_{2000}$  vs. spectral ground truth by 1.6–4.4 $\times$  over standard RGB rendering; the advantage is largest under the structured red-dominant source narrow<sub>351</sub> (0.86 vs. 3.82) and warm daylight (0.90 vs. 3.40). A 52-cell sweep over Gaussian-narrowband illuminants (13 centres  $\times$  4 FWHMs spanning the visible band) gives a mean per-cell codec advantage of 11.2 $\times$  over RGB (Supplemental Sec. 5).

*Capacity and binning baselines.* Since the learned decoder has nearly disjoint supports (Sec. 3.4), a natural question is whether plain wavelength binning at  $k=6$  would suffice. It does not. Tab. 1 evaluates the full ladder on the same Cornell-box suite (illuminant SPDs in Fig. 5): three pure binning variants (bin-average encoder, box decoder, no training), the two half-learned bases (fixed decoder shape, trained encoder), and the jointly learned codec.

Averaged over the five illuminants, every binning variant, including bins placed at our own decoder’s transitions, is about 3 $\times$  less accurate than the learned codec and worse than plain RGB. The two narrowband columns expose why: binning accuracy depends on how the source’s spectral structure falls relative to the fixed bin edges. The same adaptive bins score 1.33 when the 525 nm Gaussian lands near a bin centre and 6.69 on narrow<sub>351</sub>, whose red-dominant multi-lobed spectrum straddles several bin edges, and no placement resolves FL11’s three emission peaks (5.0–5.6 vs. our 1.8). The codec is placement-robust (1.11 and 0.86) and remains the most accurate representation even in the binning-friendly 525 nm column. Nor is the decoder shape the whole contribution: on the same equal-width boxes, learning only the encoder cuts the average error nearly in half (4.35  $\rightarrow$  2.36), and jointly learning the decoder nearly halves it again ( $\rightarrow$  1.24). The half-learned bases also expose a robustness failure the joint codec avoids: each is competitive on smooth broadband light but collapses on a different structured cell (boxes at 2.94 on the 525 nm source and 4.14 on FL11, Gaussians at 7.52 on FL11). The near-disjoint decoder is an outcome of the joint optimisation, not evidence that a box basis suffices. Fig. 6 shows the corresponding

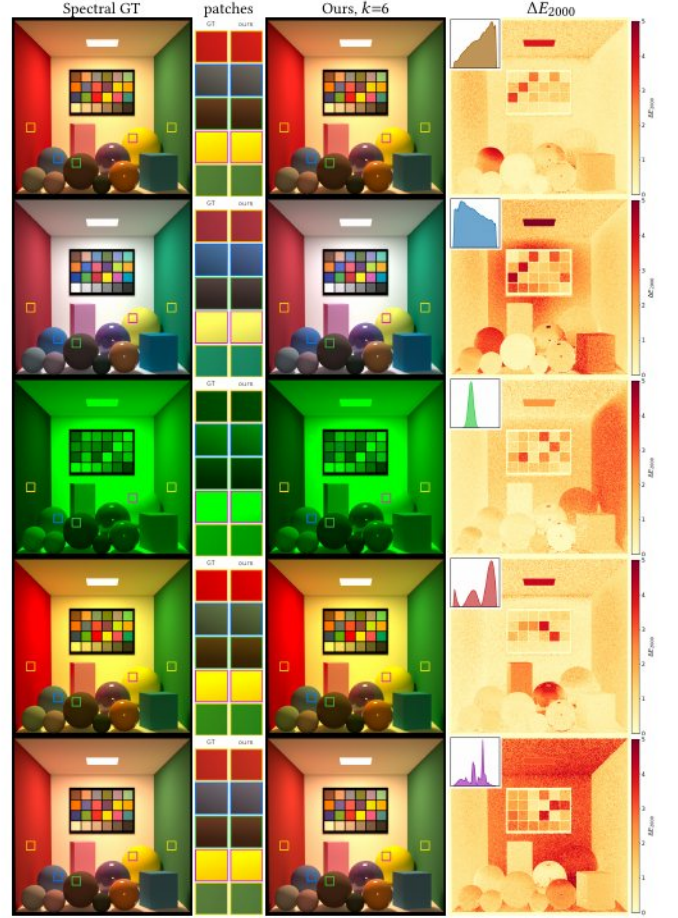


Fig. 4. **Cornell box under five illuminants.** Top to bottom: daylight 3356 K, daylight 7150 K, narrowband Gaussian at 525 nm (FWHM 40 nm), the structured red-dominant source narrow<sub>351</sub> (main peak 670 nm), and CIE FL11. The middle column shows five magnified patches (marked in the renderings; top to bottom: red wall, blue sphere, shadowed sphere, yellow sphere, green wall) as adjacent ground-truth/ours pairs for direct colour comparison. Each  $\Delta E_{2000}$  map embeds the illuminant SPD as an inset (top-left). Per-illuminant means and the RGB baseline are given in Tab. 1. 1024 spp.

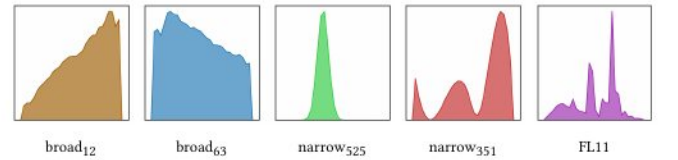


Fig. 5. **The five Cornell-box evaluation illuminants of Tab. 1 and Fig. 4,** normalised SPDs over 380–730 nm: a warm broadband (3356 K), a daylight broadband (7150 K), a 525 nm Gaussian (FWHM 40 nm), the structured red-dominant source narrow<sub>351</sub> (main peak 670 nm, secondary lobes near 540 and 410 nm), and the multi-peaked fluorescent FL11.

renders and error maps under FL11, the column where the differences are largest. PCA and NMF baselines and further per-illuminant breakdowns are in Supplemental Sec. 4.

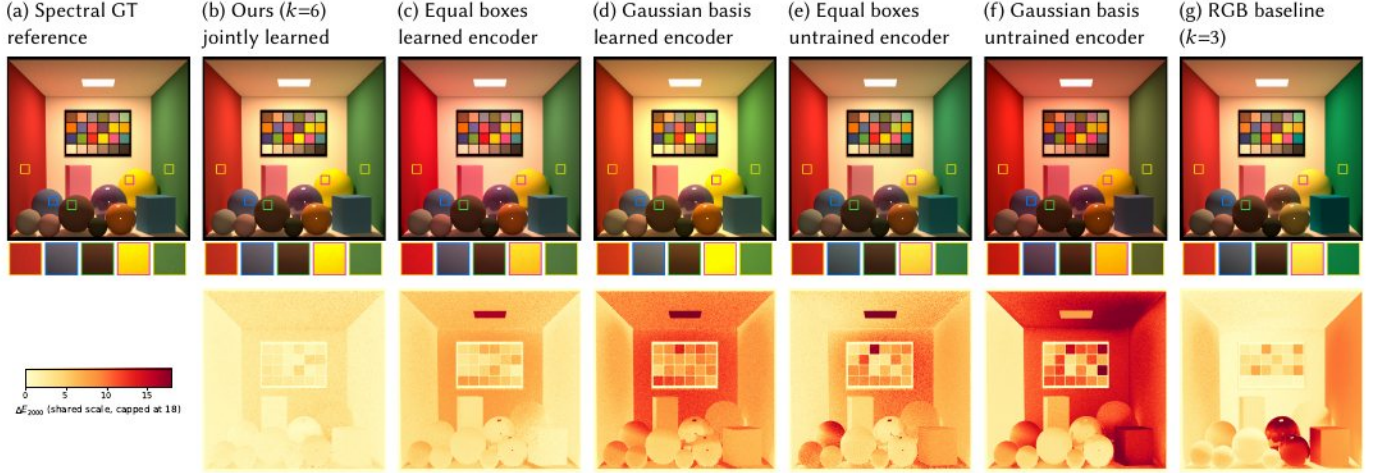


Fig. 6. **Capacity baselines on the Cornell box under FL11.** Top row: the spectral reference (a), our codec (b), the two encoder-trained bases of Tab. 1 (c, d), the same bases with untrained encoders (e, f; Supplemental Tab. 3), and the RGB baseline (g). Bottom row: each render's  $\Delta E_{2000}$  map against (a); all maps share the colour scale shown under (a), capped at  $\Delta E_{2000}=18$ . The five magnified patches under each rendering (locations as in Fig. 4) allow direct colour comparison; per-variant means are in Tab. 1 and Supplemental Tab. 3, and the FL11 SPD is in Fig. 5.

*Capacity at  $k=6$  vs.  $k=9$ .* A 13-scene cross-class sweep (the four Cornell-box illuminants, three coloured glasses, three measured conductors, and three subsurface materials) gives an aggregate  $\overline{\Delta E_{2000}}$  difference of  $-0.05$  in favour of  $k=9$ ; the gain is negligible, and per-scene differences trade in both directions rather than favouring either setting systematically (Supplemental Sec. 4.4, which also examines why the gain saturates). At 50% extra rendering cost this is not a favourable trade; we use  $k=6$  throughout.

## 6.2 Material-class evaluation

We evaluate the codec on the material classes of Sec. 4 with one scene per material, the XYZ RGB Inc. dragon (the heterogeneous SSS case uses a hand volume): coloured glass in the surface and Beer–Lambert regimes, subsurface scattering in the dipole and homogeneous regimes, measured conductors, and a heterogeneous volume. All scenes use the same  $k=6$  codec under D65; the reference is Mitsuba 0.6's spectral build. We report the object-masked mean  $\Delta E_{2000}$  against the reference after matching each candidate's mean luminance to it on the mask; the remaining materials of each class are in Supplemental Sec. 6.4.

*Coloured glass.* We evaluate five Schott filters (OG550, RG610, and GG495 longpass; BG7 and VG9 bandpass) in both regimes (Supplemental Sec. 6 renders the set as hollow vases; Fig. 7 shows OG550, BG7, and RG610): a thin transmittance filter on the dragon surface, and Beer–Lambert absorption through the solid body. Tab. 2 reports dragon-masked  $\Delta E_{2000}$  at 1024 spp.

Table 2. **Coloured glass.** Dragon-masked mean  $\Delta E_{2000}$  at 1024 spp for five Schott filters in the surface-transmittance and Beer–Lambert regimes; RGB baseline, latent pipeline, and their ratio.

| regime       | glass | RGB   | latent | gain  |
|--------------|-------|-------|--------|-------|
| surface      | OG550 | 10.52 | 1.28   | 8.22× |
|              | GG495 | 3.28  | 1.08   | 3.02× |
|              | BG7   | 6.10  | 2.06   | 2.96× |
|              | RG610 | 3.92  | 1.37   | 2.85× |
|              | VG9   | 1.74  | 1.19   | 1.46× |
| Beer–Lambert | OG550 | 10.36 | 1.72   | 6.02× |
|              | GG495 | 3.57  | 1.51   | 2.36× |
|              | BG7   | 4.14  | 1.86   | 2.23× |
|              | VG9   | 3.45  | 1.62   | 2.13× |
|              | RG610 | 2.93  | 1.80   | 1.63× |

OG550 shows the largest gain in both regimes: RGB cannot represent its sharp mid-visible cutoff. The smallest gains come from VG9 and RG610, whose spectra RGB approximates least badly.

*Subsurface scattering.* We compare the two subsurface regimes, dipole BSSRDF and homogeneous medium, side by side (Sec. 4; yellow-green dipole, skin, and wine homogeneous in Fig. 7; pale mint, rose, and potato dipole in Supplemental Sec. 6.4). Homogeneous dragons encode spectral  $\sigma_a(\lambda)$ ,  $\sigma_s(\lambda)$  (a spectral skin profile and the red-wine coefficients of the teaser scene) through the unmodified renderer's per-channel volume integrator; dipole dragons encode custom chromophore-band RGB  $\sigma$  targets (their names describe the resulting appearance), smooth-upsampled to spectra and codec-encoded. Tab. 3 reports dragon-masked  $\Delta E_{2000}$  at 1024 spp (dipole) and 4096 spp (homogeneous).

The codec advantage is largest on saturated chromophore-band materials (wine, yellow-green) and smallest on near-broadband

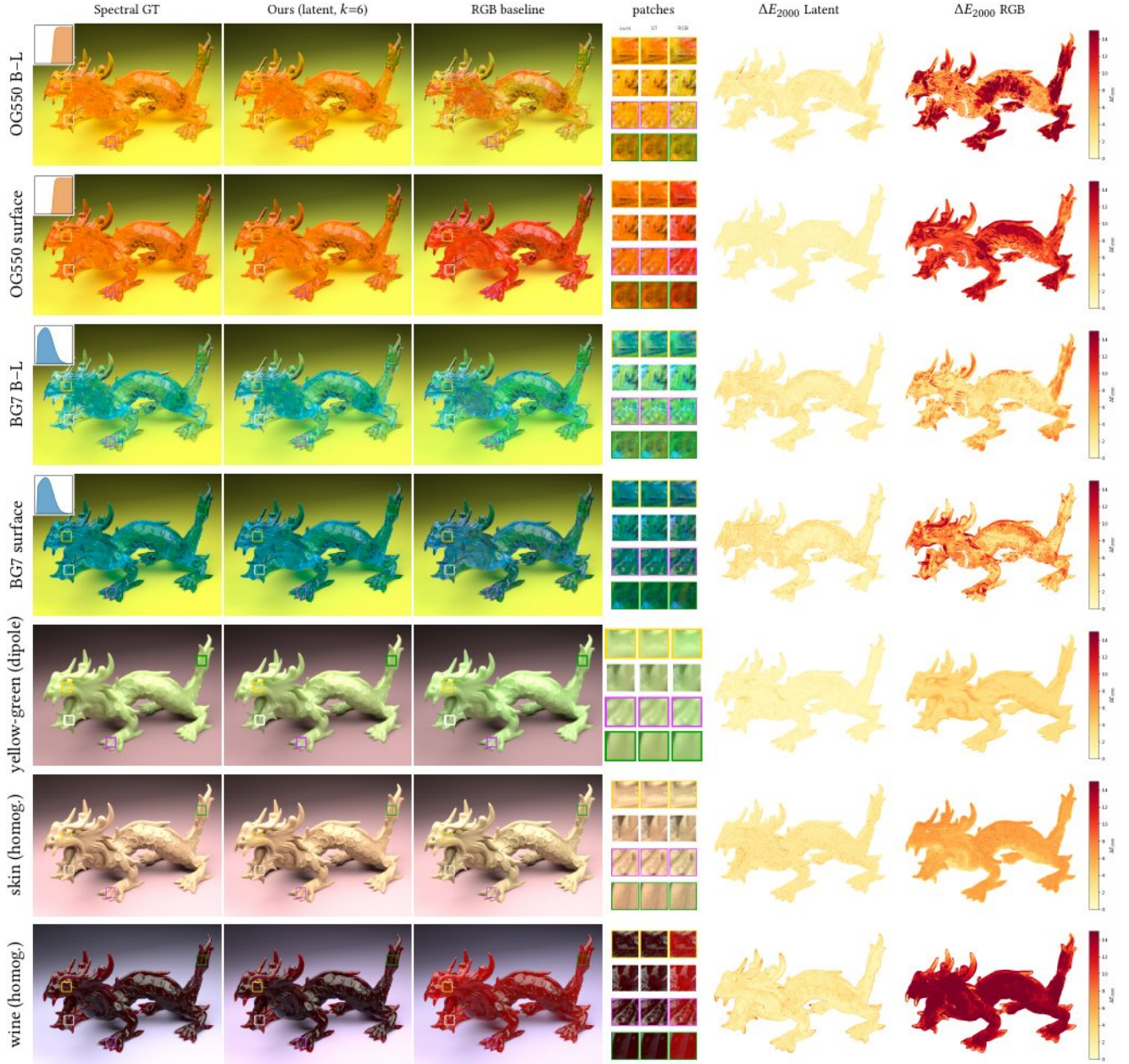


Fig. 7. **Material-class evaluation.** Seven exemplars spanning four regimes: glass under Beer–Lambert (B–L) and surface-transmittance, dipole BSSRDF, and homogeneous medium. Columns: spectral GT (Mitsuba 0.6 spectral build, CMF→sRGB); ours at  $k=6$  (two RGB passes); RGB baseline; per-pixel  $\Delta E_{2000}$  for the latent and the RGB baseline. The patches column shows four magnified regions (marked in the renderings; top to bottom: head, foreleg, forefoot, tail) as adjacent ours | GT | RGB triplets. For the glass rows the ground-truth cell embeds the filter’s normalised transmittance spectrum (inset, top-left; 400–700 nm). Per-material means are given in Sec. 6.2; the remaining five materials are shown in Supplemental Sec. 6.4. 1024 spp for the glass and dipole rows, 4096 spp for the homogeneous rows.

translucents (potato, pale mint), where the RGB baseline already captures the appearance reasonably.

*Heterogeneous subsurface scattering.* We render a per-voxel spectral hand volume through the renderer’s heterogeneous-medium

machinery (Fig. 8). The albedo is built from five anatomical features (baseline skin, freckles, knuckles, fingertips, veins) blended over the geometry; the warm features are lifted to spectra via Gitlina et al.’s

Table 3. **Subsurface scattering.** Dragon-masked mean  $\Delta E_{2000}$  at 1024 spp (dipole) and 4096 spp (homogeneous); RGB baseline, latent pipeline, and their ratio.

| mode        | material     | RGB   | latent | gain  |
|-------------|--------------|-------|--------|-------|
| dipole      | yellow-green | 3.49  | 1.09   | 3.20× |
|             | pale mint    | 1.90  | 1.52   | 1.25× |
|             | rose         | 3.97  | 2.66   | 1.49× |
|             | potato       | 2.76  | 1.22   | 2.27× |
| homogeneous | skin         | 5.61  | 2.19   | 2.56× |
|             | wine         | 14.39 | 2.19   | 6.57× |

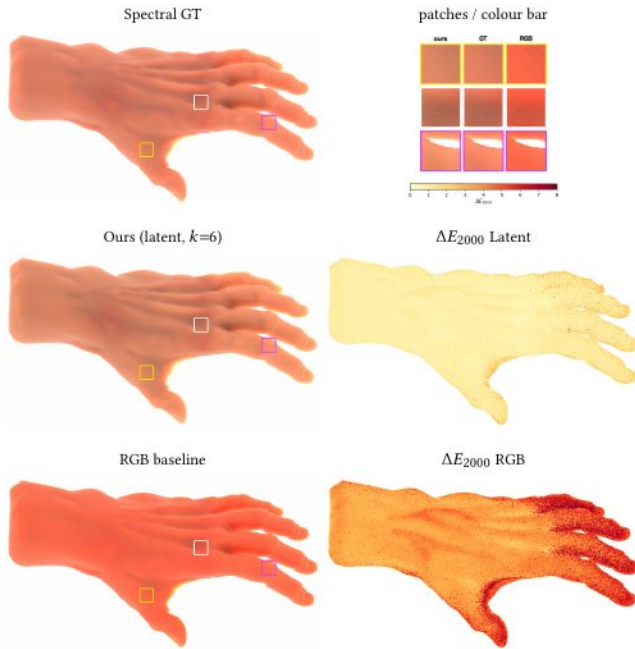


Fig. 8. **Heterogeneous subsurface scattering.** Per-voxel spectral hand volume rendered through Mitsuba's heterogeneous-medium plugin; the albedo lifts five anatomical features to spectra via Gitlina et al.'s skin reflectance LUT [2020]; construction details in Supplemental Sec. 6.3. Each rendering sits next to its own  $\Delta E_{2000}$  map; the top-right cell shows three magnified regions (palm, knuckle ridge, fingertip; marked in the renderings) as adjacent ours | GT | RGB triplets above the shared colour bar. Codec gain is 4.29× (4096 spp).

skin reflectance LUT [2020], and the vein target via a chromaticity-faithful basis fit [Guarnera et al. 2022] since it sits outside the skin gamut (construction details in Supplemental Sec. 6.3). At 4096 spp the codec attains mean  $\Delta E_{2000} = 0.93$  over the hand silhouette against 4.00 for the RGB baseline through the same volume (p95 1.89 vs. 6.62), a 4.29× gain. The advantage spans the whole visible surface: the RGB baseline's error map is substantially higher across the hand, and highest on the fingers, knuckles, and veins (Fig. 8). The heterogeneous advantage is geometry-bounded: it grows with the optical path length through coloured material, so it is most pronounced on solid bodies and muted on thin shells.

**Conductors.** We render five measured conductors (silver, gold, copper, aluminium, chromium) as a solid dragon under D65 (renders for the first three in Fig. 9). Reflectance on a conductor is a smooth but spectrally non-trivial Fresnel function of two complex IOR spectra, which the standard RGB pipeline must collapse to a single three-channel reflectance; the codec instead preserves the encoded  $\eta, \kappa$  pair and computes the Fresnel response per pass (Sec. 4; renderer bindings and a mirror-based variant in Supplemental Sec. 6.1). Tab. 4 reports dragon-masked  $\Delta E_{2000}$  at 1024 spp.

Table 4. **Conductors.** Dragon-masked mean  $\Delta E_{2000}$  at 1024 spp for five measured metals under D65; RGB baseline, latent pipeline, and their ratio.

| conductor      | RGB  | latent | gain  |
|----------------|------|--------|-------|
| Ag (silver)    | 8.43 | 2.08   | 4.06× |
| Au (gold)      | 8.98 | 0.65   | 13.9× |
| Cu (copper)    | 8.83 | 1.01   | 8.79× |
| Al (aluminium) | 8.45 | 2.11   | 4.00× |
| Cr (chromium)  | 9.27 | 2.10   | 4.42× |

Most of the RGB baseline's error is a magnitude error: collapsing the complex IOR to a three-channel Fresnel overestimates conductor reflectance by roughly a factor of two before brightness matching, with a residual hue shift for the coloured metals (gold, copper) and almost none for silver, whose spectral Fresnel is nearly neutral. The  $\Delta E_{2000}$  maps saturate across the entire dragon silhouette for the RGB baseline while the latent column stays uniformly low (Fig. 9).

**Multi-bounce stability.** On the Mitsuba matpreview test scene (Supplemental Sec. 2.4) under broadband and FL11 illumination, per-pixel error stabilises immediately: from direct lighting to one bounce the mean  $\Delta E_{2000}$  rises by at most 0.02, then remains flat through 32 bounces (the maximum tested) at 0.91/1.62 (broadband) and 1.32/1.93 (FL11) for mean/p95. The approximate Hadamard product does not accumulate at any depth; the codec self-product check in Supplemental Sec. 2.3 corroborates this.

**Approximation behaviour.** Each per-channel regime of Sec. 4 replaces an exact spectral operation with a latent one. The bounds that follow are worst-case by construction: the stress chains iterate a single encoded spectrum, whereas real paths mix different surfaces whose encoding errors partially cancel, so rendered scene means sit at the foot of the curves. Even so, the deviations are bounded and slow-growing: repeated self-products of the same encoded spectrum stay below a  $\Delta E_{2000}$  of 5 through twelve surface crossings; Beer-Lambert transmittance error rises from the codec's product floor (0.96) to 1–6 at the reference depth; per-channel conductor Fresnel is nearly angle-independent below 60°, adding about two units at grazing incidence; and the single-scatter albedo ratio stays below 6 through twenty scattering orders for smooth subsurface materials, while saturated media use the factorised encoding. The characterisation curves, evaluated exactly as the pipeline operates and overlaid with the rendered scene means of this section, are in Supplemental Sec. 6.5.

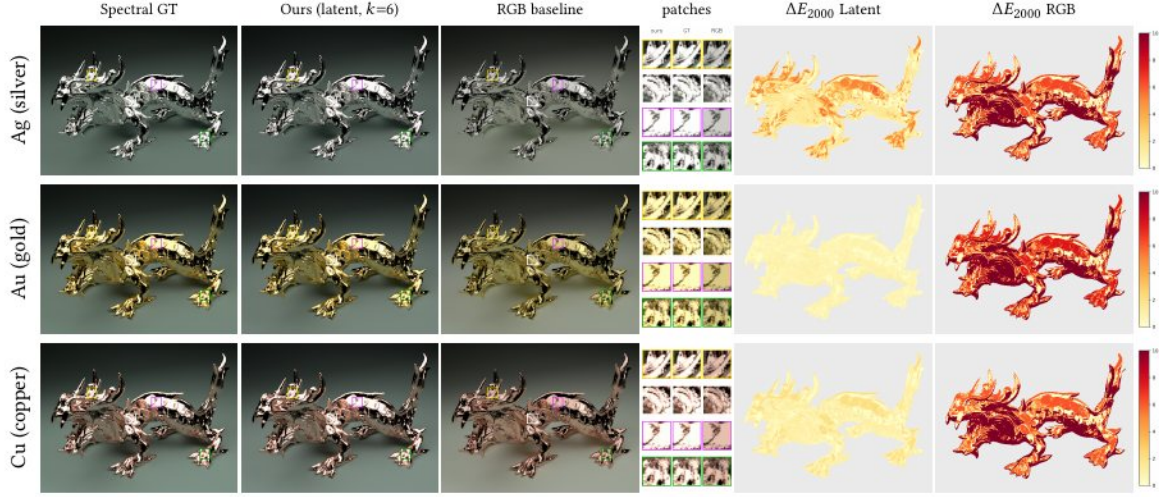


Fig. 9. **Material-class evaluation: conductors.** Three measured conductors (silver, gold, copper) rendered as a solid dragon under D65,  $k=6$  codec, dragon-masked  $\Delta E_{2000}$  versus spectral GT. Per-metal means are given in Sec. 6.2. Spectral conductor IOR is encoded via the per-component  $\eta, \kappa$  scheme of Sec. 4. The patches column shows four magnified regions (marked in the renderings) as adjacent ours | GT | RGB triplets.

Table 5. **Per-spectrum  $\Delta E_{2000}$  on 218 held-out reflectances under four illuminants** (mean / p95), grouped by what each route requires of the renderer. Bold marks the best mean and the best p95 per column *within each group*. Scaled uplift: the illuminant’s RGB uplifted with Jakob & Hanika and multiplied by D65, Mitsuba 3’s RGB environment-map convention. Guarnera et al. [2022]: per-spectrum basis fit by genetic-algorithm search, with their recommended 6-LED RGBW+ basis or their Gaussian basis.

| Method                                                                          | D65                | LED-B1             | LED-B3             | FL11               |
|---------------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|
| <b>Reflectance upsampling</b> (ground-truth illuminant)                         |                    |                    |                    |                    |
| <i>Rendered spectrally: requires a spectral renderer</i>                        |                    |                    |                    |                    |
| Jakob & Hanika                                                                  | 0.65 / 1.33        | <b>2.00 / 8.49</b> | <b>1.39 / 4.31</b> | <b>2.01 / 6.96</b> |
| Guarnera et al., LED                                                            | 0.09 / <b>0.03</b> | 5.78 / 12.62       | 3.06 / 6.09        | 2.25 / <b>5.32</b> |
| Guarnera et al., Gaussian                                                       | <b>0.08 / 0.03</b> | 5.96 / 13.48       | 5.41 / 10.08       | 4.63 / 10.77       |
| <i>Rendered through the codec: two RGB passes</i>                               |                    |                    |                    |                    |
| Jakob & Hanika                                                                  | 1.59 / 2.83        | 2.57 / 8.05        | 1.89 / 6.33        | 2.91 / 5.86        |
| Guarnera et al., LED                                                            | 2.53 / 5.45        | 4.51 / 10.70       | 3.11 / 6.60        | 3.80 / 6.92        |
| Guarnera et al., Gaussian                                                       | 1.81 / 3.66        | 4.50 / 11.32       | 3.09 / 7.73        | 5.42 / 10.46       |
| Our reflectance MLP $f^{(R)}$                                                   | <b>0.90 / 1.74</b> | <b>1.45 / 3.88</b> | <b>1.17 / 2.61</b> | <b>1.99 / 4.56</b> |
| <b>Illumination upsampling</b> (ground-truth reflectance)                       |                    |                    |                    |                    |
| <i>Rendered spectrally: requires a spectral renderer</i>                        |                    |                    |                    |                    |
| Scaled uplift (J&H×D65)                                                         | <b>0.06 / 0.11</b> | <b>1.18 / 4.11</b> | <b>1.43 / 3.58</b> | 1.48 / 4.68        |
| Guarnera et al., LED                                                            | 2.13 / 4.23        | 4.49 / 10.34       | 1.49 / <b>3.35</b> | <b>1.40 / 4.40</b> |
| Guarnera et al., Gaussian                                                       | 2.31 / 6.63        | 3.09 / 6.63        | 1.59 / 3.46        | 2.64 / 5.47        |
| <i>Rendered through the codec: two RGB passes</i>                               |                    |                    |                    |                    |
| Scaled uplift (J&H×D65)                                                         | <b>0.97 / 1.90</b> | 1.49 / 4.95        | 1.59 / 5.35        | <b>1.70 / 4.88</b> |
| Guarnera et al., LED                                                            | 3.07 / 5.73        | 4.55 / 12.12       | 1.97 / 6.17        | 1.82 / <b>4.34</b> |
| Guarnera et al., Gaussian                                                       | 2.44 / 5.82        | 3.38 / 8.52        | 2.16 / 4.66        | 2.65 / 4.88        |
| Our illumination MLP $f^{(L)}$                                                  | 1.61 / 4.10        | <b>0.96 / 2.15</b> | <b>0.86 / 3.33</b> | 1.79 / 5.38        |
| <i>Reference: no upsampling, codec on ground-truth spectra (two RGB passes)</i> |                    |                    |                    |                    |
| Codec, $k=6$                                                                    | 0.93 / 1.85        | 1.07 / 1.91        | 1.03 / 2.18        | 1.58 / 3.17        |

### 6.3 Upsampling evaluation

*Per-spectrum upsampling accuracy.* For each test reflectance  $R$  and illuminant  $L$  we compare the rendered radiance  $\hat{R}L$  (or  $R\hat{L}$ ) against

the spectral ground truth  $RL$  in CIELAB with  $L$ ’s own integrated white point. Tab. 5 reports  $\Delta E_{2000}$  across 218 held-out reflectances and four CIE illuminants for both upsampling directions. Rows are grouped by what the route asks of the renderer, because the baselines are spectral methods: each uplifts an RGB asset to a full spectrum, which a spectral renderer then transports. That is the capability our setting assumes is unavailable, so those rows are context rather than a comparison with ours. Encoding the same uplifted spectrum and rendering it in two RGB passes puts every method in the pipeline we target, and it is those groups that compare like with like. The bottom row applies the codec directly to ground-truth spectra and serves as the no-upsampling reference.

Four observations follow from Tab. 5. First, on the reflectance side the uplifting baselines are the most accurate under D65, which is their construction point: each is fitted to reproduce the target colour under the illuminant its fit is anchored to, and drifts as the illuminant departs from it, so their tails grow sharply under the structured sources while the latent path stays bounded. Second, this anchoring is most visible in the basis fit of Guarnera et al. [2022]: designed for illumination, it reproduces its calibration point almost exactly, and the same fit applied to reflectance drifts further as the illuminant departs from it. Third, on the illumination side the comparison splits by illuminant: the scaled uplift is the more accurate under D65, the two are within 0.1 under FL11, and our upsampler is the most accurate under both LED sources, where it also undercuts the ground-truth-spectra reference. Fourth, uplifting to a spectrum and then encoding composes two approximations, at a measurable cost against predicting the code directly (2.24 against 1.38 on average); the direct route exists to avoid that composition. The reflectance upsampler’s mean signed CIELAB offset matches the no-upsampling reference on every illuminant, and is smaller than the uplift-then-encode route’s on all four (Supplemental Sec. 7). Robustness beyond these four sources is probed next.

**Measured lamps.** To probe illumination robustness beyond the four CIE illuminants of Tab. 5, we evaluate the same per-spectrum pipeline on 48 measured commercial and standard lamps (LSPDD compact-fluorescent, metal-halide, and high-pressure sodium sources [LSPDD 2026] plus the CIE FL and HP series, none in the training corpus). Tab. 6 separates two questions. First, the representation itself holds up on unseen spectra: encoded directly, the codec stays close to its in-distribution accuracy across the whole population, while a pipeline with no spectral representation is several times less accurate. Second, recovery from RGB alone is a harder problem: the worst cases are the structured discharge lamps, and it is there that the routes separate, the latent upsampler having the lower error on all 48 lamps against the basis fit of Guarnera et al. and on 37 of 48 against the scaled uplift, the exceptions being fluorescent standards where the two are comparable. The same population tests the reflectance side: across the 48 lamps the direct route is more accurate than uplift-then-encode on every one.

Table 6. **Illumination robustness on unseen lamps:** mean  $\Delta E_{2000}$  over the 218 held-out reflectances, per lamp, aggregated over the 48 measured lamps as mean, median, and worst lamp. Top row: reference, the codec applied to the ground-truth lamp spectra, so no recovery is involved. Middle block: methods recovering the lamp from its RGB alone, each encoded and rendered through the codec as deployed, so all four rows share one route and one renderer requirement; bold marks the best per column within this block. Their spectral behaviour, evaluated as their authors intended, is in Tab. 5. Bottom: the RGB baseline with no spectral representation.

| Method                                                                               | 48 measured lamps |             |            |
|--------------------------------------------------------------------------------------|-------------------|-------------|------------|
|                                                                                      | mean              | median      | worst      |
| <i>Reference: codec on the ground-truth lamp spectra (two RGB passes)</i>            |                   |             |            |
| Codec, $k=6$                                                                         | 1.23              | 0.98        | 3.1        |
| <i>Lamp recovered from its RGB, then rendered through the codec (two RGB passes)</i> |                   |             |            |
| Scaled uplift (J&H×D65)                                                              | 2.98              | 2.09        | 26.3       |
| Guarnera et al., LED                                                                 | 4.28              | 3.45        | 19.8       |
| Guarnera et al., Gaussian                                                            | 3.46              | 2.95        | 9.9        |
| Our illumination MLP $f^{(L)}$                                                       | <b>2.18</b>       | <b>1.78</b> | <b>7.5</b> |
| <i>No spectral representation</i>                                                    |                   |             |            |
| RGB baseline                                                                         | 5.85              | 4.44        | 32.5       |

**Spectral environment maps.** Finally, we render the upsamplers (Sec. 5) in complete scenes: Cornell boxes lit by measured hyperspectral environment maps [Morimoto et al. 2019] (Fig. 10), pairing each baseline with the corresponding route of ours, Jakob and Hanika [2019] against  $f^{(R)}$  with a ground-truth envmap, and Guarnera et al.'s 6-LED basis [2022] against  $f^{(L)}$  with ground-truth materials. Per-cell values are in the figure. As in Tab. 5, the two baselines are rendered spectrally and carry uplift error only, while every column of ours is the deployable two-pass route and carries the codec's error as well. "Codec,  $k=6$ " applies the codec to ground-truth spectra and bounds what upsampling can achieve in these scenes. In the two maps the rendered routes are indistinguishable.

#### 6.4 Cross-renderer integration

To verify that the codec deploys without renderer modification, we run it through four production frameworks: Blender Cycles

(open-source path tracer; broadband and narrowband conditions), Autodesk Maya with Arnold (offline production), KeyShot (envmap-driven product visualisation), and Unreal Engine 5 (real-time rasteriser with Lumen global illumination). In each pipeline the latent code is treated as RGB material textures and light SPDs over two passes, and the decoder is a matrix multiply on the pass outputs.

Spectral ground truth is not natively available in these commercial pipelines, so we construct one per renderer: every material is uplifted to a spectrum with Jakob and Hanika [2019] (the reference is not built with our codec), light sources use their known SPDs, and each scene is rendered in 16 RGB-packed passes that reconstruct the full spectrum, stored as linear floating-point data with per-renderer response and exposure calibration (KeyShot and Cycles at 128 spp, Arnold at AA level 3, Unreal via the Movie Render Queue; configurations in Supplemental Sec. 9). Against this reference we evaluate three routes per scene (Figs. 11–14): the codec applied to the same spectral assets in two RGB passes, which isolates the codec's own error in each pipeline; the full RGB-asset route through our upsamplers, which adds the recovery cost; and the renderer's plain RGB baseline.

Across the thirteen scene and lighting cases the codec's own error stays below  $2.8 \Delta E_{2000}$  in every renderer and lighting condition, the RGB-asset route adds a bounded upsampling premium, largest under Unreal's deliberately metameric lighting, and the plain RGB baseline is the least accurate of the three in twelve of the thirteen cases. The exception is a nearly flat broadband source in Maya, smoother across the visible band than D65, under which all three routes fall within  $2 \Delta E_{2000}$  of the reference. Per-case means and 95th percentiles are in Supplemental Sec. 9. KeyShot is evaluated both under its environment map and under uniform-colour environment lights carrying the same known SPDs as the Maya cases; the RGB pipelines are the renderers' own linear outputs, brightness-matched to the reference.

A 120-frame animated sequence in Blender Cycles confirms temporal stability: no flicker or chromatic drift across either broadband or narrowband condition (per-renderer configurations and the supplemental video are in Supplemental Sec. 9).

#### 6.5 Performance vs. hero wavelength sampling

At  $k=6$  our method requires two RGB render passes, so its per-frame cost is about twice an RGB render by construction and comparable to a modern spectral renderer with hero wavelength sampling [Wilkie et al. 2014]. We benchmark end-to-end, including codec encode and decode, against Mitsuba 3's spectral path tracer at matched resolution and spp (Fig. 15, Tab. 7; full configurations in Supplemental Sec. 8).

Table 7. **Wall-clock ratio, latent (two RGB passes) over Mitsuba 3 hero wavelength sampling,** across the spp and resolution sweep of each scene.

| scene                          | frame time bound by        | latent / spectral |
|--------------------------------|----------------------------|-------------------|
| Cornell box, 15 configurations | shading                    | 0.94–1.10         |
| Glass of Water                 | shading                    | 1.13–1.22         |
| Veatch MIS                     | shading                    | 1.13–1.29         |
| Grey & White Room              | traversal                  | 1.75–1.81         |
| + three FL11 lamps             | traversal, heavier spectra | 1.63–1.73         |

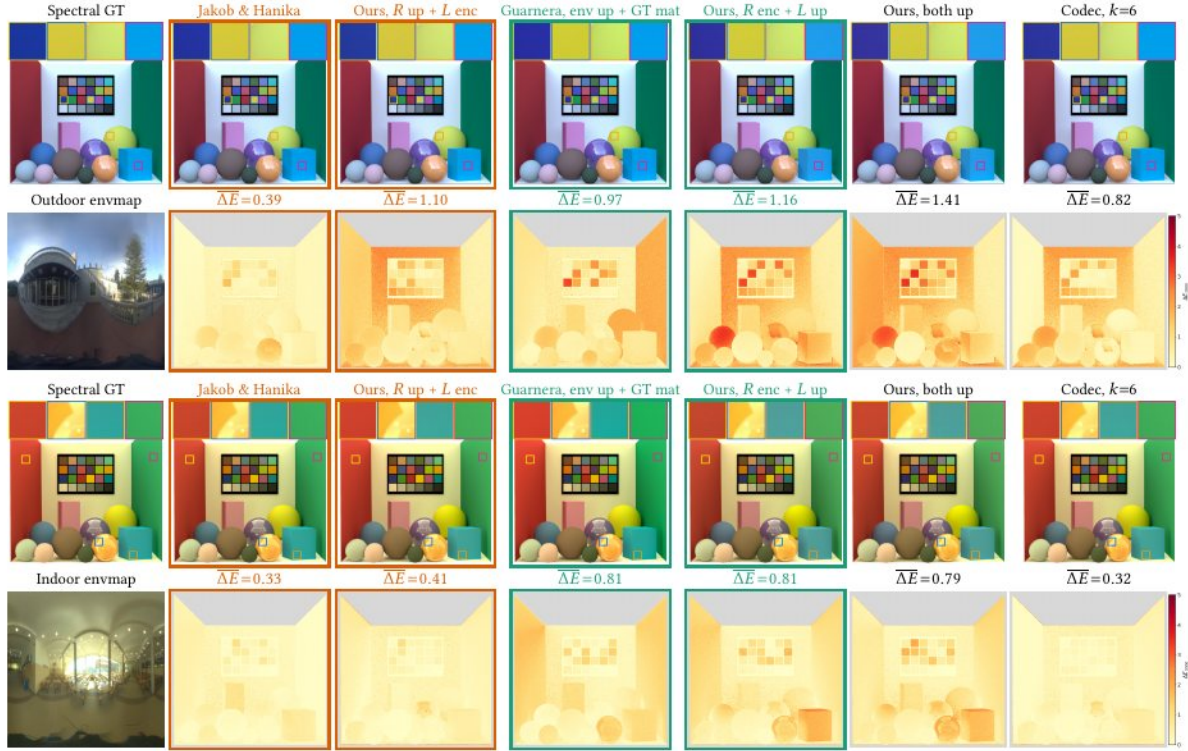


Fig. 10. **Latent upsampling on Cornell-box scenes lit by spectral environment maps.** Outdoor (rows 1–2) and indoor (rows 3–4); odd rows render, even rows  $\Delta E_{2000}$  with envmap thumbnail leftmost (sky pixels excluded from means). Coloured borders group each baseline with the corresponding route of ours: **reflectance upsampling with GT envmap** (Jakob & Hanika [2019] vs. ours, R up); **illumination upsampling with GT materials** (Guarnera et al. [2022] vs. ours, L up). The two baselines are rendered spectrally and carry uplift error only; every column of ours is the deployable two-pass RGB route and carries the codec’s error as well. “Ours, both up” upsamples both sides from RGB; “Ours,  $k=6$ ” applies the codec to GT spectra. 512 spp; per-cell  $\Delta E_{2000}$  above each error map. Hyperspectral environment maps from Morimoto et al. [2019].

On the Cornell box the two-pass pipeline is at parity: within  $\pm 10\%$  on 14 of 15 resolution and spp configurations, median +5%, one at +20%. On heavier scenes the ratio depends on how much of the frame time is wavelength-dependent shading rather than geometry traversal: near parity where shading dominates, up to  $1.8\times$  on a textured interior dominated by traversal, and adding spectral load to that same interior narrows the gap. Reconstruction quality across the same spp and resolution sweep is reported in Supplemental Sec. 8. Decoding is a single matrix multiply on the framebuffer and stays below 0.5% of frame time at 1024 spp and above.

## 7 Conclusion

We introduced a learned linear non-negative spectral codec that compresses spectra into  $k$ -dimensional spectral codes, exactly preserving scaling and addition and approximately preserving the per-channel product on which renderer shading operates. At equal  $k$ , the learned basis is consistently more accurate than binning: on the Cornell box every binning variant is about  $3\times$  less accurate, learning the encoder over fixed boxes nearly halves the error, and jointly learning the decoder nearly halves it again (Sec. 6.1). At  $k=6$ , two unmodified RGB passes through any path tracer reconstruct a spectral image to a mean  $\Delta E_{2000}$  between 0.7 and 2.7 (median 1.8) of

the spectral ground truth across coloured glass, conductors, dipole and homogeneous subsurface scattering, and a per-voxel heterogeneous subsurface-scattering hand (Sec. 6.2); end-to-end wall-clock time is at near parity with Mitsuba 3’s hero wavelength sampling on shading-bound scenes and about  $1.8\times$  on a traversal-bound interior (Sec. 6.5). Legacy RGB assets enter the same pipeline through small XYZ-conditioned upsampling networks that add a bounded premium over the no-upsampling reference (Sec. 6.3). Deployment is renderer-agnostic: we drove four production engines, Blender Cycles, Maya Arnold, KeyShot, and Unreal Engine 5, through the same two-pass interface unmodified, recovering colour information the RGB pipeline collapses. The renderer never changes; only the meaning of its three channels does. This is efficient *spectral rendering without a spectral renderer*.

*Scope.* The codec preserves only what reduces to per-channel arithmetic. Wavelength-shifting phenomena couple distinct wavelengths during transport and cannot be captured by a per-channel  $\odot$  regardless of  $k$ : fluorescence, phosphorescence, and the chromatic dispersion that arises from a wavelength-dependent index of refraction. BSDF non-linearities that escape per-channel arithmetic (e.g. a coated layer whose internal Fresnel contains  $\exp(-\sigma d)$  along a wavelength-dependent path) similarly fall outside the supported

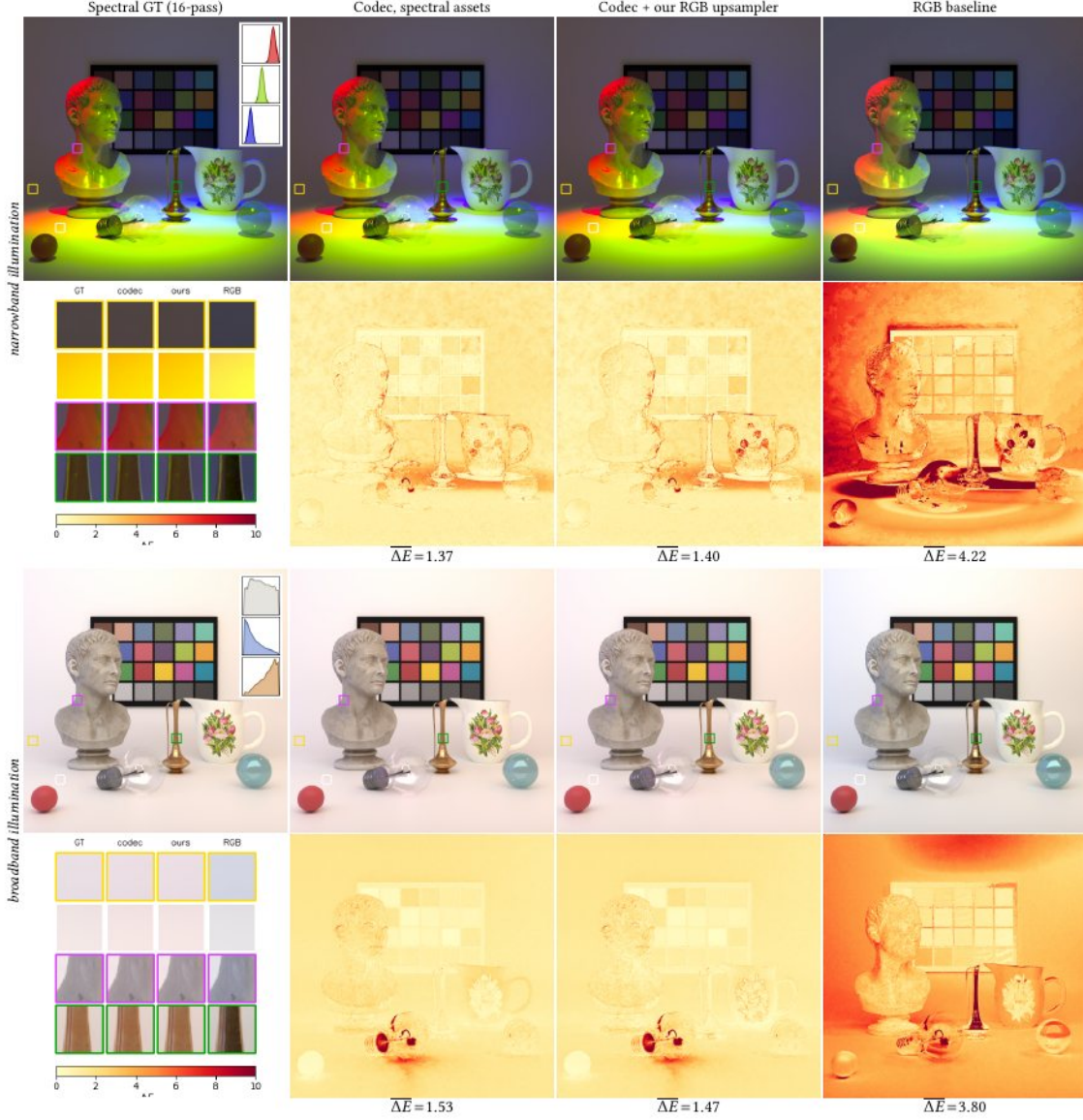


Fig. 11. **Cross-renderer evaluation: Blender Cycles** (open-source path tracer), two lighting conditions. For each lighting condition, top row: the 16-pass spectral reference; the codec applied to the same spectral assets (two RGB passes); the full RGB-asset route through our upsamplers; and the renderer's plain RGB baseline. Bottom row: four magnified patches (marked in the renderings) as adjacent GT | codec | ours | RGB swatches with the shared colour bar, then the per-pixel  $\Delta E_{2000}$  of the three non-reference columns against the reference (means below; scale capped at 10). RGB-baseline renders are brightness-matched to the reference. The inset in the reference cell shows the illumination: one panel per light SPD, or the environment map where one is used.

regime; Sec. 4 enumerates the cases that are inside it. Layered materials, whose cross-layer coupling is non-linear in the spectra, are a natural target for future work under the same constraint. Residual error is largest under extremely peaked illuminants and after many indirect interactions, but Sec. 6.2 shows it stabilises with bounce count rather than diverging. The advantage is concentrated where an RGB pipeline discards spectral information. On spectrally bland media and under smooth broadband light the RGB baseline loses little, and the codec's own reconstruction floor of one to two  $\Delta E_{2000}$

units can match or exceed it, as in the broadband Maya case of Sec. 6.4. For effects outside the regime, the codec is a pre-encoder for the linear part of the pipeline and would need to be paired with a renderer-side handler for the non-linear residual.

Three practical boundaries follow from the construction. The codec operates on the 400–700 nm band; energy outside it contributes little to CMF-integrated colour, and extending the wavelength grid requires only retraining on a wider corpus, not an architectural change. The two passes must remain linear until decode:

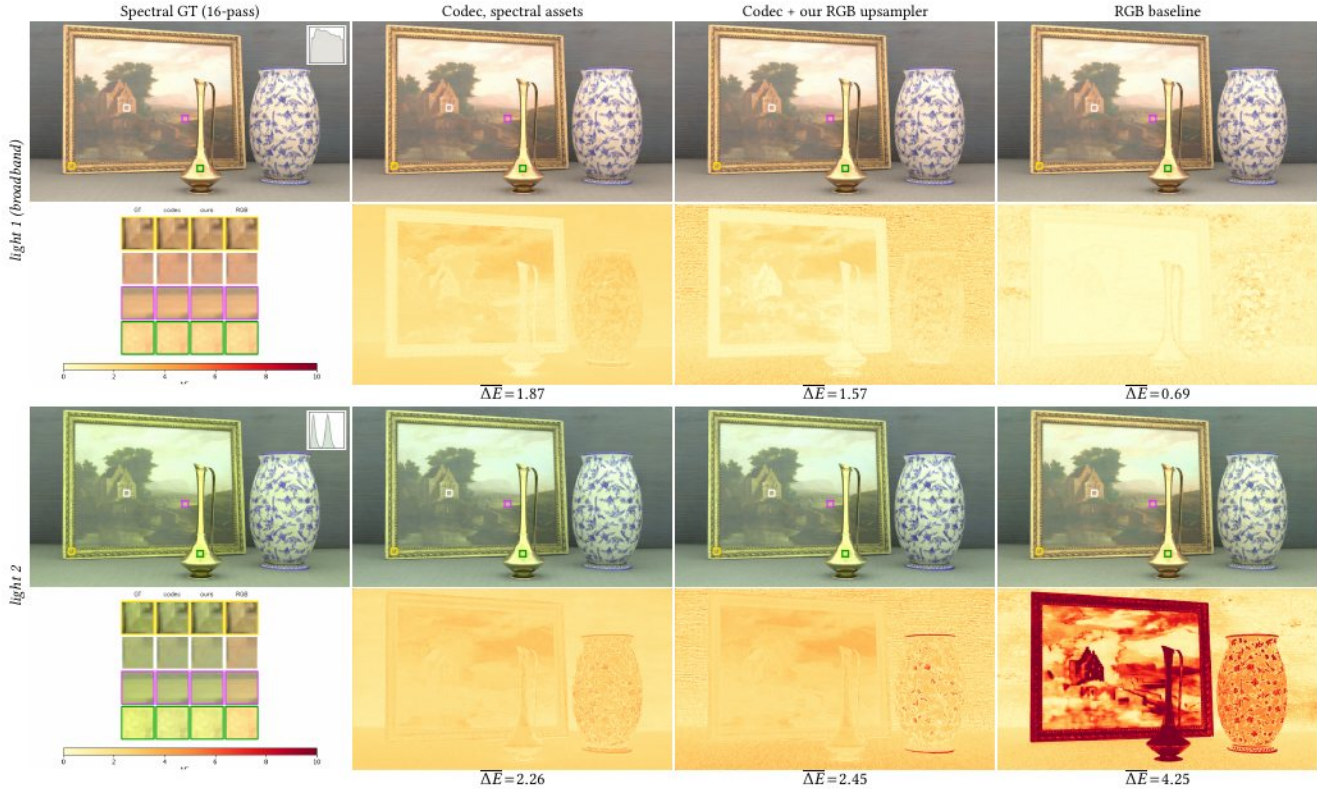


Fig. 12. **Cross-renderer evaluation: Maya Arnold** (offline production renderer), two lighting conditions: light 1 is a nearly flat broadband source, smoother than D65 and the case an RGB pipeline handles best; light 2 is a spectrally structured source. Layout as in Fig. 11. Under light 1 all three routes lie within  $2\Delta E_{2000}$  of the reference and the RGB baseline is the most accurate (0.69); under light 2 the RGB baseline rises to 4.25 while both codec routes stay near 2.3.

tone mapping or other non-linear frame operations on the per-pass buffers before decoding would break the product algebra, so decoding must precede any such post-processing. Finally, the encoder weights wavelengths non-uniformly, so near-monochromatic sources at adjacent wavelengths can encode at different magnitudes; accuracy on measured and held-out lamps is unaffected, and the effect is confined to near-monochromatic light (Supplemental Sec. 5).

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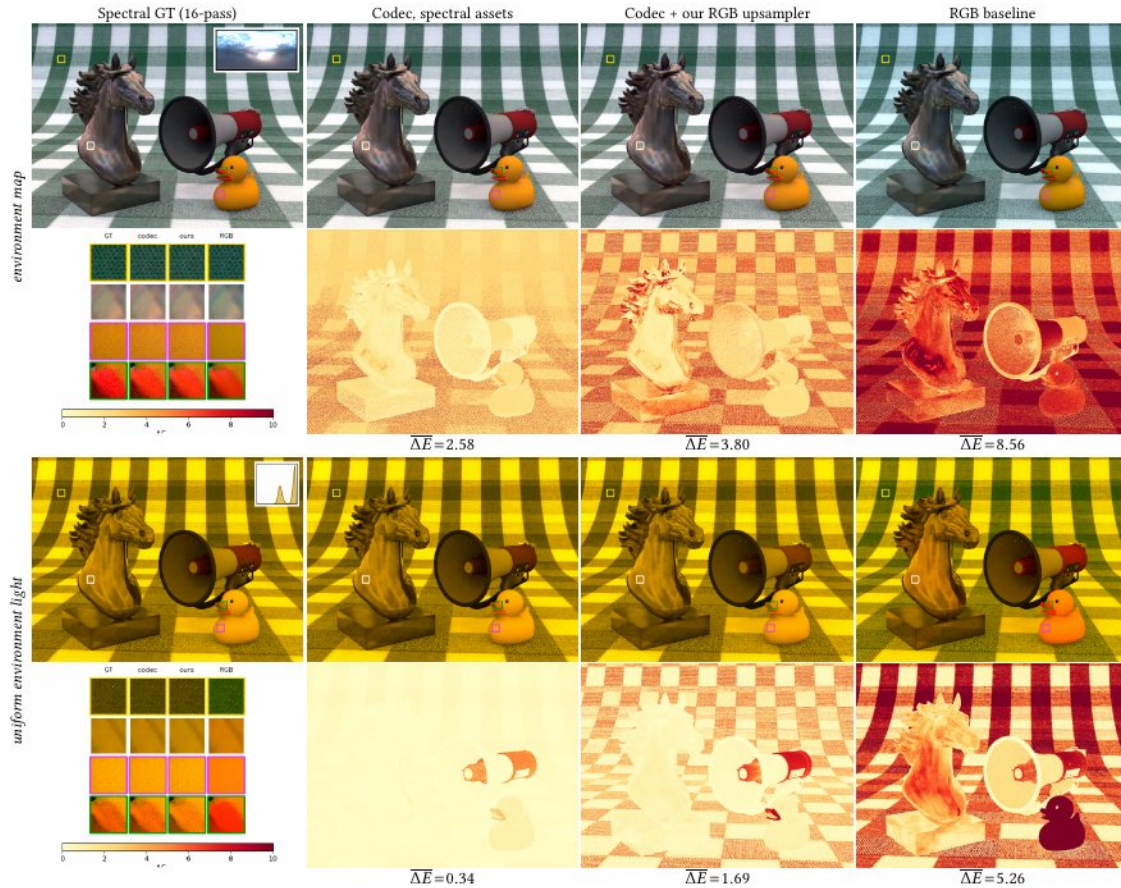


Fig. 13. **Cross-renderer evaluation: KeyShot** (product visualisation), two lighting conditions. Layout as in Fig. 11. For the environment-map condition the reference sky spectrum is the Jakob & Hanika uplift of the RGB map, which is an 8-bit image, so the reference sky is spectrally flat wherever the map is clipped.

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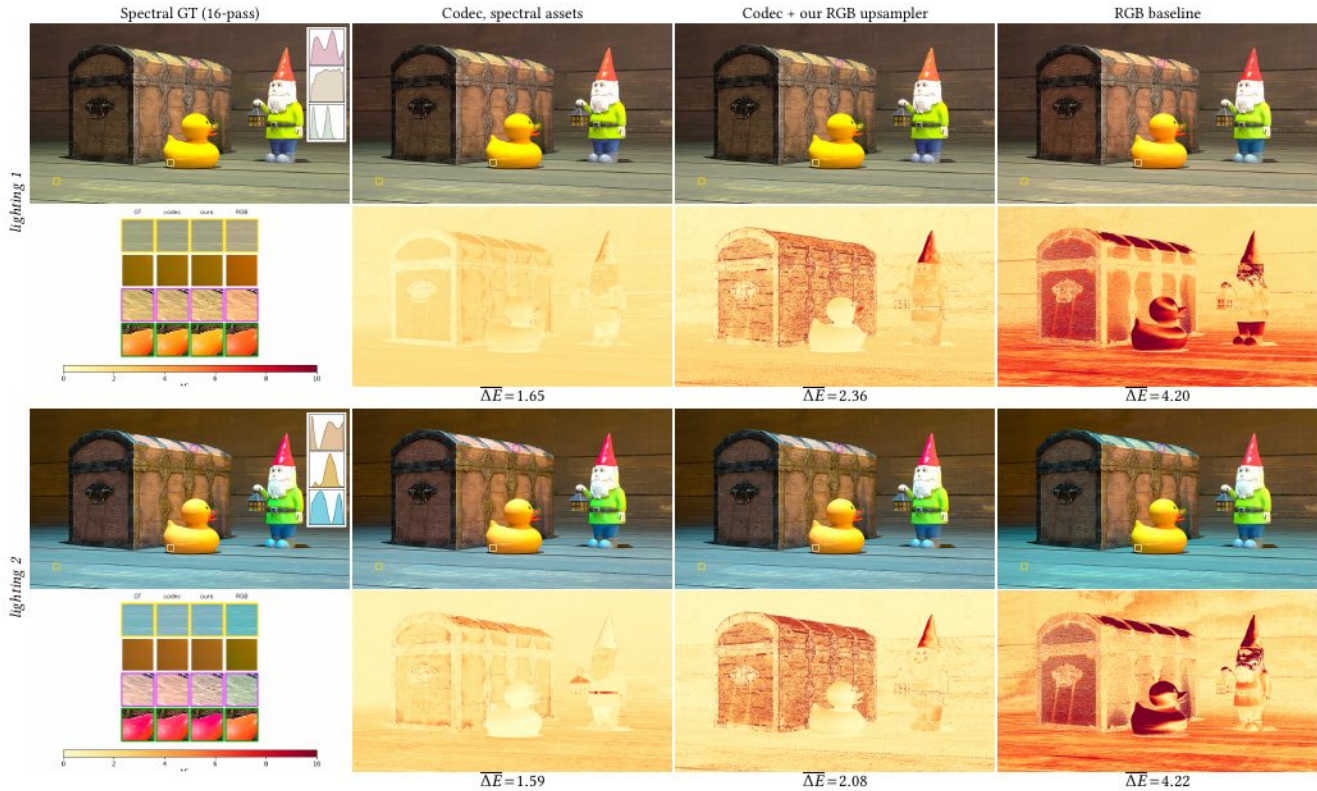


Fig. 14. **Cross-renderer evaluation: Unreal Engine 5** (real-time rasteriser with Lumen), two lighting conditions. Layout as in Fig. 11.



Fig. 15. **Timing scenes** (spectral reference, 1920×1080, 1024 spp). Left to right: Glass of Water (aXel, CC0) and Veach MIS (B. Bitterli, CC0), both shading-bound; the Grey & White Room (Wig42, CC BY 3.0), a textured interior whose frame time is dominated by traversal. The controlled-load variant adds three FL11 ceiling lamps outside the view.

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